

Gabriel Smithline

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Personal Profile

I am passionate about exploring the intersection between multi-agent systems, economics & computation, reinforcement learning, and mathematical economics & finance. I hold United States and Polish citizenship.

Personal Website: <https://gsmithline.github.io/>

Education

University of Michigan Ann Arbor

Ann Arbor, MI

PhD in Computer Science and Engineering

August 2024 - Expected May 2028

- Member of the Strategic Reasoning Group in the AI Lab
- **Advisor:** Michael P. Wellman
- Currently researching game theory & cooperative AI, bargaining games, and cooperation in competitive settings

Tufts University

Medford, MA

MS in Computer Science

May 2022 - May 2024

- Graduated summa cum laude while working full time
- Thesis: Deception as a Defensive Strategy in Stackelberg Security Games
- Degree Focuses: Theoretical Computer Science, Machine Learning, and AI

Lafayette College

Easton, PA

BA Mathematical Economics, and Statistics-Data Science

August 2017 - May 2021

- Data Education and Feminism Scholar at Lafayette and Beyond (DEFLAB)
- 4 year Division 1 Student-Athlete on the men's lacrosse team committing 30 plus hours a week to activities on top of school
- USILA All American
- Thesis: A Statistical and Game Theoretic Approach to a Better Housing Market

Industry Work Experience

Tufts University

Medford, MA

Machine Learning Course Assistant

August 2023 - May 2024

- Helped run office hours, grade assignments, run recitations, answer questions for students, and generally support them in the machine learning course.

Jefferies

New York, New York

Quantitative Researcher

May 2023 - January 2024

- Worked to model risk for fixed income trading as well as helping to build out the banks e-trading and algorithmic trading business. Specifically I helped to automate VaR and Monte-Carlo VaR simulations as well as help develop software to price certain fixed income securities. I also developed internal software to run VaR calculations on full fixed income trading data.
- The results of my work helped speed up systematic risk calculations by **50%**
- Developed statistical and mathematical models for pricing fixed income assets

Capital One

Washington, DC

Software Engineer, Enterprise Data and Machine Learning

August 2021 - December 2022

- Scaled and designed API to handle more than 52,000 transactions per second with less than 7 ms response time.
- Enabled canary deployments, identified/remediated bottlenecks in code, built features to track lineage of all data flowing through the API.
- Helped data scientists and ml engineers run models and perform inference in low latency environments for credit fraud detection.
- Became Certified AWS Solutions Architect.

Additional Projects and Research

Humanity Unleashed: AI Safety, Value Elicitation, and Incentive Design

Remote

Project Technical Lead

October 2024 - Present

- Utilizing mathematics, game theory, mechanism design, and social choice theory to develop a rigorous methodology for safe and aligned AI.
- Mentoring undergraduates and masters students as they implement our work.
- I'm being mentored by researchers at OpenAI and MILA on this project, as well as working with other PhD students.

Improving Multi-Agent Strategies through Learning

Shalimar, Florida

Happ Arnold Summer Fellow, AI and Control Theory Research Lab Eglin Air force Base

May 2024 - August 2024

- Spent the summer researching multiagent systems, differential game theory, pursuit-evasion, Geometric Deep Learning, and Graph Theory. The paper for this is under review.
- Hosted by Dr. Scott Nivison

Complexity Theory, Algorithmic Game Theory, and Security Research

Remote

Tufts University

August 2023 - May 2024

- Worked to develop models of deception as a defensive strategy against attackers using various optimization techniques, Game Theory, AI, Mechanism Design, and other concepts from Computer Science.

Algorithms, Law, and Policy Working Group Member

Remote

Equity and Access in Algorithms, Mechanisms, and Optimization (EEAMO)

September 2023 - Present

- Member of the Algorithms, Law, and Policy group which is working on research, implementation, and advocacy projects.